



The ESO Astroquery module: a Jupyter Notebooks walkthrough

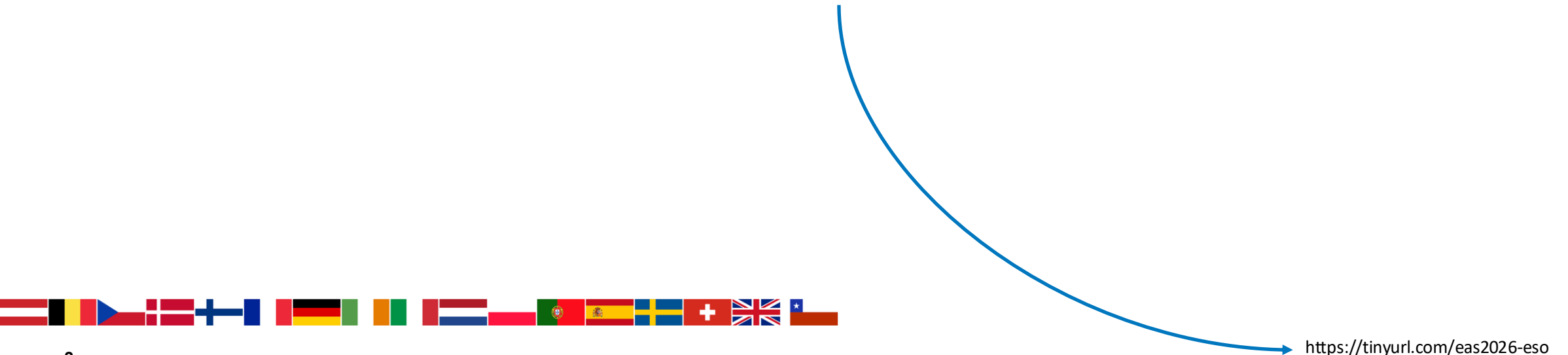
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DE & DS, ESO Back-end Operations Department



Hands-on session

Please download the slides from below

https://github.com/eso/astroquery_examples/blob/main/assets/eas2026-ucc-eso-astroquery.pdf





What is Astroquery?

Astroquery enables end-to-end analyses

 “Astroquery is a set of tools for querying astronomical web forms and databases.”

 **Documentation:** astroquery.readthedocs.io

 **Code and issue tracker:** <https://github.com/astropy/astroquery>

 **Paper:** Ginsburg, Sipőcz, Brasseur et al 2019.

Available Services

The following modules have been completed using a common API:

- ALMA Queries ([astroquery.alma](#))
- Atomic Line List ([astroquery.atomic](#))
- Besancon Queries ([astroquery.besancon](#))
- CADZ ([astroquery.cadc](#))
- CASDA Queries ([astroquery.casda](#))
- Cologne Database for Molecular Spectroscopy (CDMS) Queries ([astroquery.linealists.cdms](#))
- ESA EUCLID Archive ([astroquery.esa.euclid](#))
- ESA Herschel Science Archive ([astroquery.esa.hsa](#))
- ESA HST Archive ([astroquery.esa.hubble](#))
- ESA Integral Science Legacy Archive (ISLA) ([astroquery.esa.integral](#))
- ESA ISO Archive ([astroquery.esa.iso](#))
- ESA JWST Archive ([astroquery.esa.jwst](#))
- ESA XMM-Newton Archive ([astroquery.esa.xmm_newton](#))
- ESASky Queries ([astroquery.esasky](#))
- ESO Queries ([astroquery.eso](#))
- FIRST Queries ([astroquery.image_cutouts.first](#))
- Gaia TAP+ ([astroquery.gaia](#))
- GAMA Queries ([astroquery.gama](#))
- Gemini Queries ([astroquery.gemini](#))
- HEASARC Queries ([astroquery.heasarc](#))
- HiPS2fits Service ([astroquery.hips2fits](#))



- HITRAN Queries ([astroquery.hitran](#))
- IRSA Moving Object Search Tool ([astroquery.ipac.irsa.most](#))
- IRSA Dust Extinction Service Queries ([astroquery.ipac.irsa.irsa_dust](#))
- IRSA Image Server program interface (IBE) Queries ([astroquery.ipac.irsa.ibe](#))
- IRSA Queries ([astroquery.ipac.irsa](#))
- JPL Spectroscopy Queries ([astroquery.jplspec](#))
- MAGPIS Queries ([astroquery.magpis](#))
- MAST Queries ([astroquery.mast](#))
- CDS MOC Service ([astroquery.mocserver](#))
- Minor Planet Center Queries ([astroquery.mpc/astroquery.solarsystem.MPC](#))
- NASA ADS Queries ([astroquery.nasa_ads](#))
- NED Queries ([astroquery.ipac.ned](#))
- NIST Queries ([astroquery.nist](#))
- NVAS Queries ([astroquery.nvas](#))
- SIMBAD Queries ([astroquery.simbad](#))
- Skyview Queries ([astroquery.skyview](#))
- Splatalogue Queries ([astroquery.splatalogue](#))
- SVO Filter Profile Service Queries ([astroquery.svo_fps](#))
- UKIDSS Queries ([astroquery.ukidss](#))
- Vamdc Queries ([astroquery.vamdc](#))
- VizieR Queries ([astroquery.vizieR](#))
- VO Simple Cone Search ([astroquery.vo_conesearch](#))
- VSA Queries ([astroquery.vsa](#))
- xMatch Queries ([astroquery.xmatch](#))

The ESO module delivers complex datasets quickly

 “Simple python functions to script your access to the ESO archive via TAP.”

 ESO astroquery fork: <https://github.com/eso/astroquery>

 ESO example notebooks: https://github.com/eso/astroquery_examples

- *The ESO astroquery module allows users to search for raw and reduced data, retrieve metadata, access catalogues, and download data products. Proprietary access is also supported.*

WDB (“Old”)

- Limited query interfaces with predefined queries
- Users can only enter constraints on predefined queries
- No direct query language
- Users cannot modify underlying queries
- Catalogues not accessible
- Simplified access for specific, common queries

TAP (“New”)

- Full read-only access to ESO databases
- Users can write their own custom queries
- Uses ADQL 2.0 (SQL-based, VO standard)
- Users fully control query structure
- Catalogues accessible via dedicated interface
- Advanced, flexible querying for complex data retrieval

- *The **new** ESO module enables faster, larger and more complex dataset analyses by using the the Table Access Protocol (**TAP**), an International Virtual Observatory Alliance (**IVOA**) standard.*

Functionality

- List instruments and data releases.
- Instrument-specific and instrument-independent raw data search.
- Search across / by data releases.
- Automatic calibration selection (raw).
- Get extended FITS header information.
- Download datasets by their identifiers.
- Authenticated access to your assets.
- Run custom / complex ADQL queries.
- Retrieve atmospheric conditions data.

→ Automate all the above ←





**Where to get the new
ESO astroquery module and examples?**



Get the ESO module and examples from GitHub

Overview Repositories 6 Projects 1 Packages Teams People 32

ESO
20 followers International http://www.eso.org

Popular repositories

- astroquery_examples** (Public)
Jupyter notebooks for querying, accessing, and analyzing data from the ESO Science Archive using astroquery.eso. Includes examples for La Silla, Paranal (VLT), APEX, and ALMA datasets.
8 stars 3 forks
- astroquery** (Public)
Forked from [astro/astroquery](#)
Functions and classes to access online data resources. Maintainers: @keflavich and @bsipocz and @ceb8
Python 1 fork
- edps-doc** (Public)
TeX
- homebrew-pipelines** (Public)
A homebrew tap for ESO instrument pipelines
Ruby 3 stars 1 fork
- homebrew-test-pipelines** (Public)
A homebrew tap for ESO instrument pipelines (test version)
Ruby
- pyvo_examples** (Public)
Jupyter notebooks and scripts for querying, accessing, and analyzing data from the ESO Science Archive (mainly) using pyvo. Includes examples for La Silla, Paranal (VLT), APEX, and ALMA datasets.
Jupyter Notebook

astroquery

- Fork of the official astroquery repository
- Latest ESO features not yet part of an official astroquery release



Get the ESO module and examples from GitHub

astroquery Public
forked from [astroquery/astroquery](#)

Edit Pins Watch 1 Fork 1 Star 0

develop-catalogues 7 Branches 0 Tags

Go to file Add file Code

This branch is 30 commits ahead of and 53 commits behind [astroquery/astroquery:main](#) Contribute Sync fork

	ashleythomasbarnes	Remove temporary ROW_LIMIT handling	08882d8 · 2 months ago	10,831 Commits
📁 .github		Bump codecov/codecov-action from 5.5.2 to 6.0.0 in the a...	3 months ago	
📁 astroquery		Remove temporary ROW_LIMIT handling	2 months ago	
📁 docs		Merge pull request astroquery#3566 from bsipocz/DOC_fix_...	2 months ago	
📁 licenses		Moving license to root directory [skip ci]	9 years ago	
📄 .gitignore		made (hopefully) all changes from code review	4 years ago	
📄 .gitmodules		MAINT: removing AH	4 months ago	
📄 .mailmap		MAINT: update mailmap	9 months ago	
📄 .readthedocs.yaml		RTD: installing optional dependencies for RTD builds, too	4 months ago	
📄 CHANGES.rst		Update CHANGES.rst	2 months ago	
📄 CITATION		CITATION file should be in the source to be distributed to ...	7 years ago	

About

Functions and classes to access online data resources. Maintainers: @keflavich and @bsipocz and @ceb8

[astroquery.readthedocs.org/en/lat...](#)

Readme BSD 3-Clause license Contributing Cite this repository Activity Custom properties 0 stars 1 watching 1 fork Report repository

Releases

No releases published [Create a new release](#)

astroquery

- **Important** for this example session we need the catalogue development version of astroquery

pip install

"git+https://github.com/eso/astroquery.git@develop-catalogues"



- **astroquery**
 - Fork of the official astroquery repository
 - Latest ESO features not yet part of an official astroquery release



Get the ESO module and examples from GitHub

Overview Repositories 6 Projects 1 Packages Teams People 32

ESO 20 followers International http://www.eso.org Unfollow

Popular repositories

astroquery_examples Public

Jupyter notebooks for querying, accessing, and analyzing data from the ESO Science Archive using astroquery.eso. Includes examples for La Silla, Paranal (VLT), APEX, and ALMA datasets.

☆ 8 🍴 3

astroquery Public

Forked from [astro/astroquery](#)

Functions and classes to access online data resources. Maintainers: @keflavich and @bsipocz and @ceb8

Python 🍴 1

edps-doc Public

TeX

homebrew-pipelines Public

A homebrew tap for ESO instrument pipelines

Ruby ☆ 3 🍴 1

homebrew-test-pipelines Public

A homebrew tap for ESO instrument pipelines (test version)

Ruby

pyvo_examples Public

Jupyter notebooks and scripts for querying, accessing, and analyzing data from the ESO Science Archive (mainly) using pyvo. Includes examples for La Silla, Paranal (VLT), APEX, and ALMA datasets.

Jupyter Notebook

View as: Public

You are viewing the README and pinned repositories as a public user.

People

View all

Top languages

Python Ruby TeX Jupyter Notebook

Most used topics

astronomy eso eso-archive examples notebooks-jupyter

pyvo_examples

Jupyter notebooks with tutorials and real application examples making use of pyvo module.



Get the ESO module and examples from GitHub

astroquery_examples

- Jupyter notebooks with tutorials and real application examples

astroquery_examples (Public)

main 3 Branches 0 Tags

Go to file Add file Code

ashleythomasbarnes include more description of previews and updated .pngs c254bba · last month 77 Commits

File	Commit	Time
.binder	Update runtime.txt	4 months ago
assets	Upload eas2025 slides	last year
examples	include more description of previews and updated .pngs	last month
.gitignore	Moved around files...	10 months ago
LICENSE	Initial commit - README + LICENSE	last year
README.md	updated catalog files wording	9 months ago

README MIT license

ESO Science Archive - Jupyter Notebooks

⚠ Work in Progress — Updated: 22 August 2025

These notebooks are **under active development**. While we aim for accuracy, we **cannot guarantee** that everything in this repository currently works as intended.

Some functionality depends on features of `astroquery` that are **only available in the [ESO astroquery fork](#)**. Until these changes are merged into a stable release, please install `astroquery` directly from that branch to ensure compatibility.

Features, functions, and outputs may change without notice. Please report any issues via GitHub.

Introduction

This repository contains a suite of Jupyter notebooks designed to demonstrate how to explore, query, and retrieve data from the [ESO Science Archive](#) using Python. The primary interface used is the `astroquery.eso` module, which is part of the broader [astroquery](#) package developed under the Astropy ecosystem.

Contributors 4

- ashleythomasbarnes Ashley Bar...
- almicol Alberto Micol
- gmella Guillaume Mella





Get the ESO module and examples from GitHub

Name	Last commit message	Last commit date
..		
advanced	update dawn analysis with comments	9 months ago
case_studies	include more description of previews and updated .pngs	last month
simple	updated catalog files wording	9 months ago
README.md	Updated to highlights -> case_studies	10 months ago

Introduction to basic functionality

- Query the archive for raw* and reduced data
- Perform a simple cone search
- Obtain extended information on data products
- Download datasets from the archive

*Either instrument-specific or generic

Tutorials on basic functionality

- APEX products - reduced, raw, quick looks
- Searching via ADQL (SQL-like syntax)
- Downloading datasets by identifier
- Searching a source by name

Real life examples and use cases

- Find the closest observation to a given time
- Investigate ESO data using Aladin
- Download raw data for a reduced data product
- Examine Spectral Data as function of time



Get the ESO module and examples from GitHub

main ▾ astroquery_examples / examples / simple /

T Add file ▾

ashleythomasbarnes Add instrument-specific query example 9351e0d · 8 minutes ago History

Name	Last commit message	Last commit date
..		
00_quickstart.ipynb	Add instrument-specific query example	8 minutes ago
01_authentication.ipynb	checks for simple examples	last month
02_observations_query_reduced_by_position.ipynb	checks for simple examples	last month
03_observations_query_raw_by_position.ipynb	checks for simple examples	last month
04_observations_query_by_program_id.ipynb	checks for simple examples	last month
05_observations_query_observation_apex.ipynb	checks for simple examples	last month
06_observations_query_tap.ipynb	Add TAP cone, polygon and spatial examples	last month
07_observations_download_data.ipynb	checks for simple examples	last month
08_catalogs_query_with_constraints.ipynb	checks for simple examples	last month
09_catalogs_query_by_position.ipynb	checks for simple examples	last month
10_catalogs_query_tap.ipynb	updates to query_tap	last month
README.md	Replace introduction notebook with quickstart	16 minutes ago



Take a test drive

– Hands-on session –

Take a test drive on the cloud



Scroll down the github page and click here – **may take a little time to load up, so be patient...**

The image is a screenshot of a web browser displaying the README for the astroquery ESO Science Archive interface. The page has a light blue header with a 'README' tab and a 'MIT license' link. The main content is titled 'Introduction' and describes the repository's purpose: to demonstrate how to explore, query, and retrieve data from the ESO Science Archive using Python. It mentions the primary interface is the `astroquery.eso` module, part of the broader `astroquery` package. The text then explains that the notebooks walk through both introductory and advanced examples, covering a wide range of scientific use cases from downloading raw data to analyzing spectra. A large orange curved line highlights the 'Examples' section. This section is titled 'Examples' and states that all notebooks live under the `examples/` directory and are grouped by purpose. A code block shows the directory structure: `examples/` contains `simple/` (Minimal, single-feature recipes), `advanced/` (Scenario-driven workflows), and `case_studies/` (Curated case studies). Below this, it says 'See detailed indexes inside each folder:' and lists three links: [examples/simple/README.md](#), [examples/advanced/README.md](#), and [examples/case_studies/README.md](#). The 'Quick Start' section features a 'launch binder' button, which is highlighted by the orange line. Below this, it says 'To get started, you can load and initialize the ESO archive interface directly in your notebook:' and provides a code snippet:

```
from astroquery.eso import Eso # Import the ESO module from astroquery
eso = Eso()                  # Create an instance of the ESO class
eso                           # Display the class instance (e.g. check login status, default
```

The screenshot shows a Jupyter Notebook environment for the ESO Science Archive. The interface includes a file explorer on the left, a main workspace with a title bar, and a bottom panel for code execution. The notebook content is as follows:

ESO Science Archive Jupyter Notebooks

Quick Start

This notebook gives a compact first workflow for using `astroquery.eso` with the ESO Science Archive. It shows how to initialize the ESO Interface, run a small positional search for raw and reduced observations, optionally download the matching products, and query a published catalogue table.

Note: Catalogue and TAP-backed archive access require an `astroquery.eso` version with the newer TAP support, such as `astroquery` 0.4.12 or later. If a query method used below is unavailable in your environment, update `astroquery` before continuing.

The later notebooks in this folder provide more detailed examples for authentication, raw observations, reduced products, TAP/ADQL queries, catalogue searches, and data downloads.

Importing and basic usage of `astroquery.eso`

We start by checking the installed `astroquery` version and creating an `Eso` instance. The `Eso` object stores the archive configuration used by the query and download helpers.

```

import astroquery
print(f"astroquery version: {astroquery.__version__}") # check the version of astroquery

[1] ✓ 1.5s Python

... astroquery version: 0.1.dev10831+g88882d882

from astroquery.eso import Eso
from astro.coordinates import SkyCoord
import astropy.units as u

[2] ✓ 1.6s Python

```

Take a test drive on the cloud



The screenshot shows the GitHub repository page for 'astroquery_examples'. The 'Clone' dropdown menu is open, showing options for 'Local' and 'Codespaces'. Under 'Local', there is a 'Clone' button and a list of cloning methods: 'HTTPS', 'SSH', and 'GitHub CLI'. The 'HTTPS' method is selected, showing the URL 'https://github.com/eso/astroquery_examples'. Below the cloning methods, there are options to 'Open with GitHub Desktop' and 'Download ZIP'. An orange arrow points from the 'Clone' menu to a text box on the right.

Local Codespaces

Clone

HTTPS SSH GitHub CLI

https://github.com/eso/astroquery_examples

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

You can also download locally and run the notebooks on your machine – **but make sure you have the correct version of astroquery installed**



Take a test drive on the cloud

astroquery_examples Public

main 3 Branches 0 Tags

Go to file T Add file <> Code

Local Codespaces

Codespaces
Your workspaces in the cloud

No codespaces
You don't have any codespaces with this repository checked out

[Create codespace on main](#)

[Learn more about codespaces...](#)

Codespace usage for this repository is paid for by ashleythomasbarnes.

ashleythomasbarnes include more description of previews and updates

- .binder Update runtime.txt
- assets Upload eas2025 sli
- examples include more descri
- .gitignore Moved around files.
- LICENSE Initial commit - REA
- README.md updated catalog file

README MIT license

ESO Science Archive - Jupyter Notebooks

For this example we will use github codespaces – which you can find by clicking on “code” and then “create codespace on main” - be patient this can take a little time to load...

Don't worry about this – its free if you have a github account!

Thank you!

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@ESO Astronomy



@esoastronomy



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european-southern-observatory



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